Employee Data Analysis Report

Data Science Internship | Starta Company
April 2026

Table of Contents

1. Introduction

This report presents a comprehensive analysis of a sample employee dataset from a company. The analysis was conducted using Python with the Pandas, Matplotlib, and Seaborn libraries. The dataset contains 1,000 employee records across 14 attributes including department, salary, age, gender, and ethnicity.

2. Data Exploration

The dataset was loaded into a Pandas DataFrame and explored to understand its structure and content.

2.1 Dataset Overview

Total Rows	1,000
Total Columns	14
Columns	EEID, Full Name, Job Title, Department, Business Unit, Gender, Ethnicity, Age, Hire Date, Annual Salary, Bonus %, Country, City, Exit Date

3. Data Cleaning

Data cleaning was performed to handle missing values and ensure data quality before analysis.

3.1 Missing Values

Column	Missing Values
Full Name	2
Job Title	1
Department	2
Gender	1
Ethnicity	7
Age	6
Annual Salary	11
Bonus %	8

Missing values were filled using the median for numerical columns and "Unknown" for categorical columns.

4. Data Modification

The first 5 rows were modified with updated values as follows:

Row	Column Modified	New Value
Row 0	Job Title	Senior Data Analyst
Row 1	Department	IT

Row 2	Annual Salary	$95,000
Row 3	City	New York
Row 4	Bonus %	10%

5. Data Analysis

5.1 Employee with Highest Salary

The employee with the highest annual salary is Raelynn Rios with an annual salary of $258,498, working in the IT department.

5.2 Average Age and Salary by Department

The following table shows the average age and average annual salary grouped by department:

Department	Average Age	Average Salary
Accounting	43.66	$123,146.95
Engineering	45.67	$109,053.10
Finance	45.21	$122,418.86
Human Resources	44.44	$117,907.09
IT	44.37	$97,942.35
Marketing	43.22	$129,663.03
Sales	43.61	$111,120.64

5.3 Max Age, Min Age, and Median Salary by Department & Ethnicity

A detailed analysis was performed by grouping the data by both Department and Ethnicity. For each group, the maximum age, minimum age, and median salary were calculated.

6. Data Visualizations

The following figures present visual insights derived from the employee dataset.

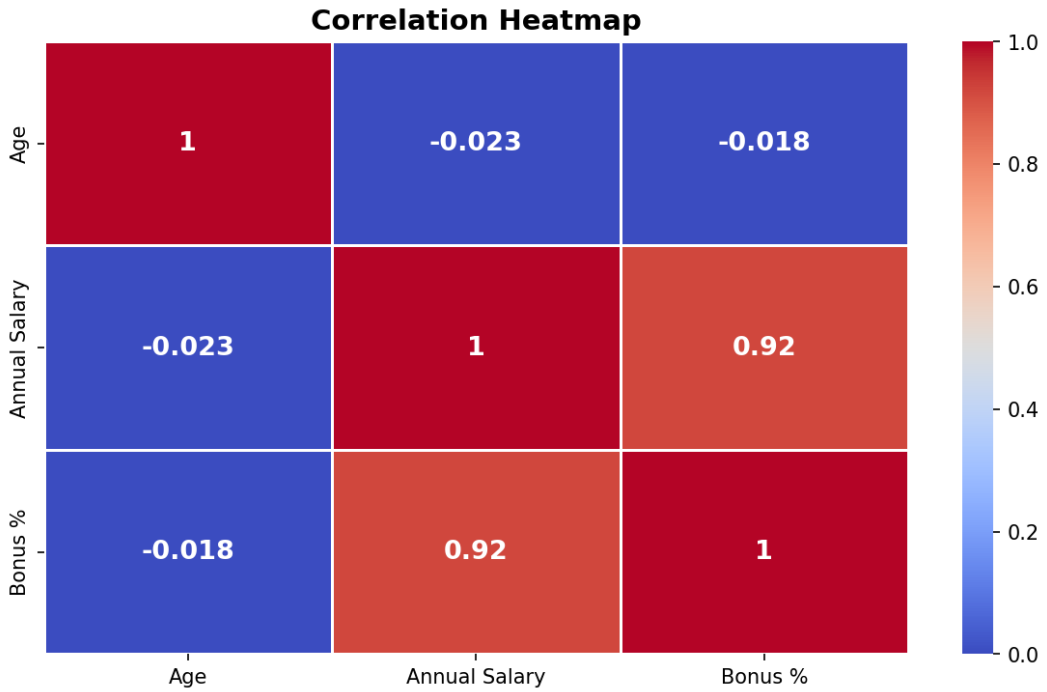

Figure 1: Correlation Heatmap – Age vs Annual Salary = -0.023 | Age vs Bonus % = -0.018 | Annual Salary vs Bonus % = 0.92

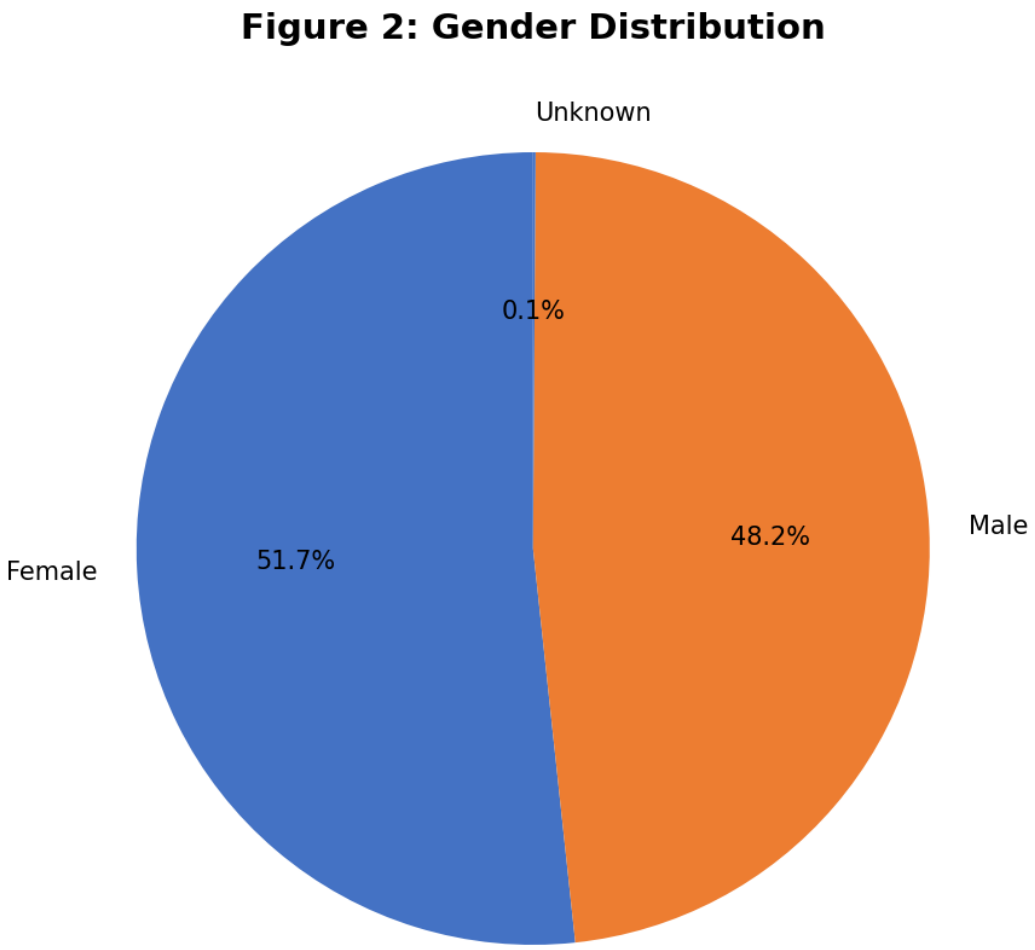

Figure 2: Gender Distribution – 51.8% Female and 48.2% Male employees.

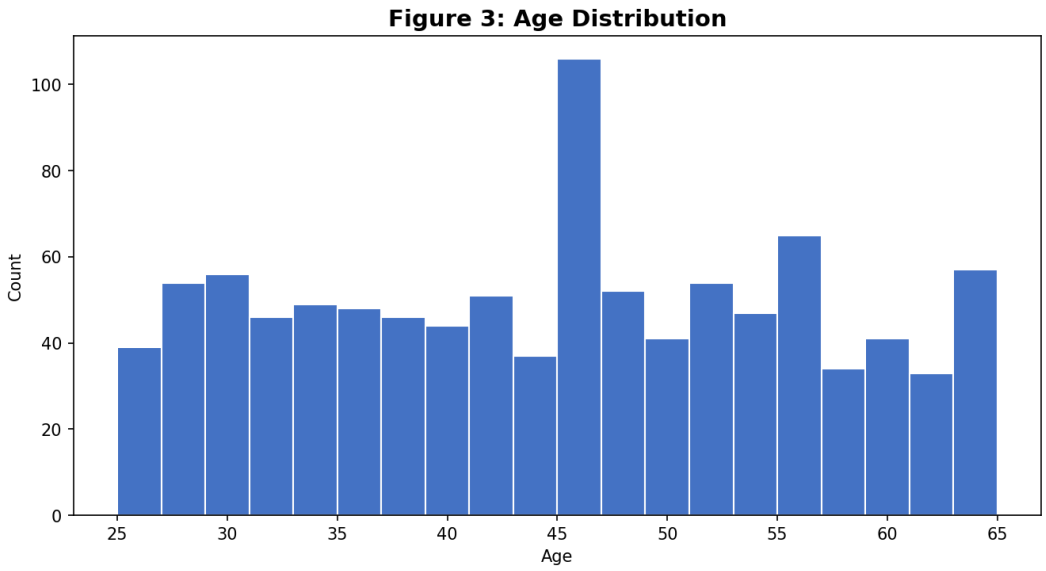

Figure 3: Age Distribution – Most employees fall between ages 25 and 65.

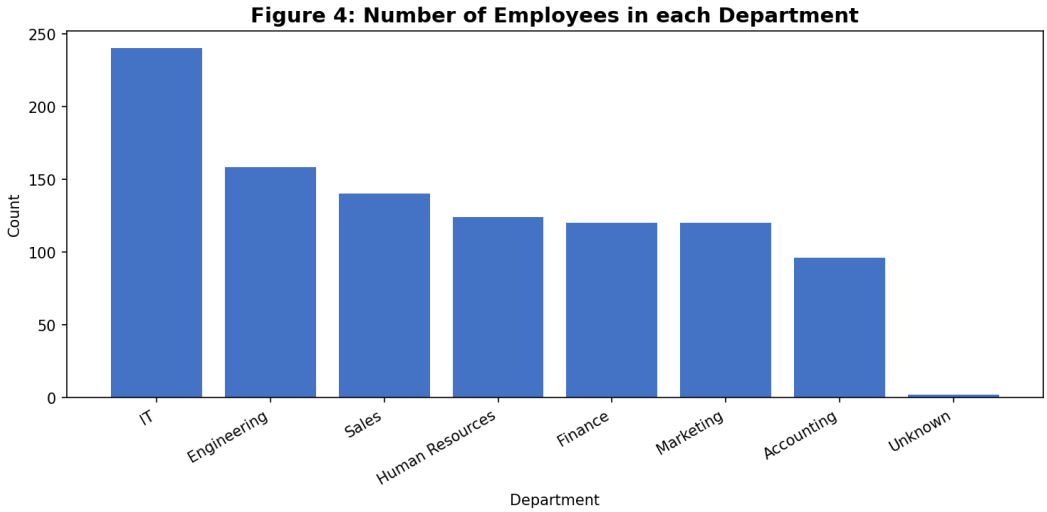

Figure 4: Number of Employees in each Department – IT has the highest number of employees.

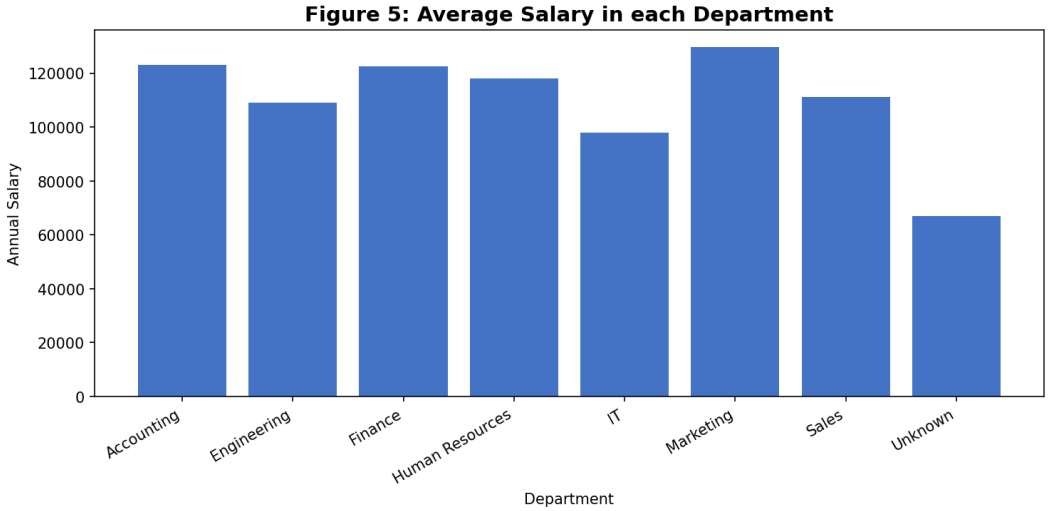

Figure 5: Average Salary in each Department – Marketing has the highest average salary.

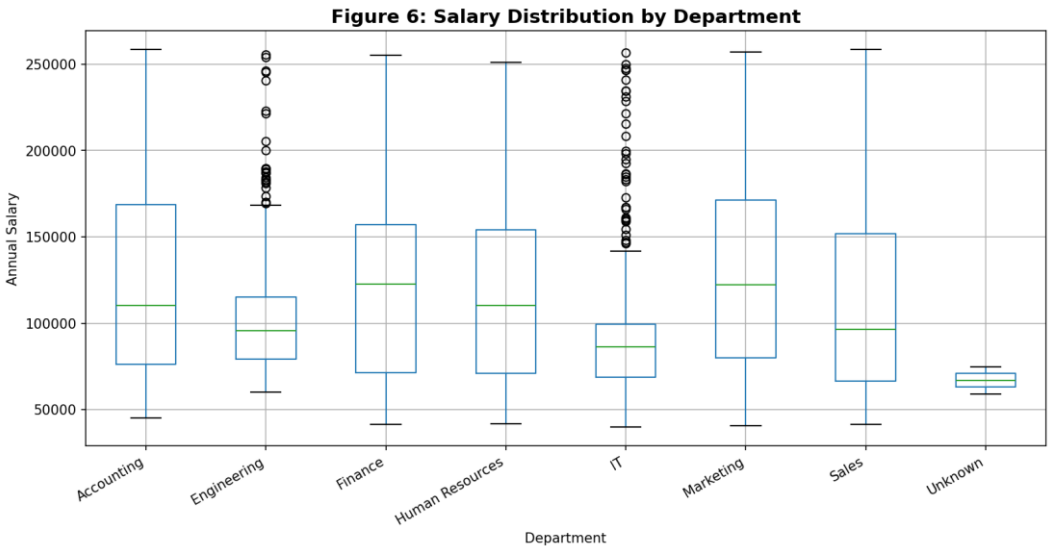

Figure 6: Salary Distribution by Department – Box plot showing salary spread and outliers per department.

7. Conclusion

This analysis provided valuable insights into the employee dataset. Key findings include:

- The dataset contains 1,000 employees across 7 departments.
- Marketing has the highest average salary ($129,663), while IT has the lowest ($97,942).
- The highest-paid employee is Raelynn Rios with an annual salary of $258,498.
- Gender distribution is nearly equal: 51.8% Female and 48.2% Male.
- Annual Salary and Bonus % have a strong positive correlation of 0.92.
- Most employees are between 25 and 65 years old.
- The cleaned dataset was saved as Employee_Clean.xlsx for further use.